Lecture 1 - Introduction to Euclidean path integrals and lattice scalar field theory

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Abstract

In this lecture we begin by motivating the study of lattice field theory first introducing the Euclidean path integral for quantum mechanics, promoting the discussion to field theory and discussing how it can be used to compute expectation values and correlation functions. We then motivate the discretization of spacetime and how it helps regulate the path integral. Finally, we introduce the lattice scalar field theory as a simple example to illustrate how to put a theory on a discrete spacetime lattice. The first half of these notes are based on [1] while the second half on [2].

1 Overview

The Standard Model of particle physics is a quantum field theory that describes three of the four fundamental forces of nature — the electromagnetic, weak and strong interactions. While we have made a lot of progress in studying the electromagnetic and weak forces using tools like perturbation theory, the strong force requires a non-perturbative treatment. Lattice quantum chromodynamics (lattice QCD) provides a *first-principles*, non-perturbative treatment of QCD that has been very successful in its study of the strong force for the last ~ 50 years. The two basic ingredients to understand lattice field theory are: (i) Euclidean path integral formalism, and (ii) learning how to put theories on a discrete spacetime grid. In this first lecture we will try to motivate these two ideas starting from the real-time path integral of a simple quantum mechanical system, and building our way to a lattice scalar field theory. Once we warm up to these ideas, we will be in a position to put more complicated theories like gauge fields and fermions on the lattice.

2 Real time path integral

In this section we will briefly discuss the real-time path integral and its properties - as applied to quantum mechanics. We will see that the real-time path integral is a powerful tool for computing transition amplitudes and correlation functions, but it suffers from convergence issues due to the oscillatory nature of the integrand. This motivates the *Wick rotation* to *Euclidean time*, which we will discuss in the next section.

We begin by considering the Schrödinger equation for a quantum system with Hamiltonian H:

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle. \tag{1}$$

Assuming the Hamiltonian to be time-independent, the formal solution to this equation can be written as

$$|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle. \tag{2}$$

Using this we can introduce the notion of a *time-evolution* operator given by,

$$U(t', t) \equiv e^{-iH(t'-t)/\hbar}, \tag{3}$$

whose matrix elements between an initial state $|x_i, t_i\rangle$ and a final state $|x_f, t_f\rangle$ give us *transition amplitude* or the *propagator*, expressed as

$$\langle x_f|U(t_f - t_i)|x_i\rangle = \langle x_f|e^{-iH(t_f-t_i)/\hbar}|x_i\rangle. \tag{4}$$

By inserting a complete set of energy eigenstates and computing $\langle x|U(t_f - t_i)|x\rangle$, i.e., the amplitude of finding the particle at the same place after a certain interval of time - one realizes that the propagator is related to the energy spectrum by a Fourier transform[1]. If one could diagonalize the Hamiltonian easily, one could directly evaluate Eq. (4) in terms of the energy eigenstates of H - this is a very challenging task and only for very trivial theories or non-trivial theories with very small system sizes is the diagonalization of H possible. Something that we can do is recognize that the Hamiltonian for most systems of interest is made up of (non-commuting) parts that are easier to treat in their own bases. For the example we will use below - that of a non-relativistic quantum mechanical particle - the Hamiltonian becomes the kinetic and potential energy given by,

$$H = \frac{p^2}{2m} + V(x). \tag{5}$$

Since we already know the action of $\frac{p^2}{2m}$ on momentum eigenstates and $V(x)$ on position eigenstates, we can start to think about solving Eq. (4) by splitting the terms in the exponential and inserting a complete set of momentum and position eigenstates between the two separated exponentials. This would have worked in one step if $[p^2, V(x)] = 0$ - but this is unfortunately not the case. We need to consider the Baker-Campbell-Hausdorff formula to split the exponential of the sum of two non-commuting operators into a product of exponentials. This is given by,

$$e^{A+B} = e^A e^B e^{-\frac{1}{2}[A,B]} \dots \tag{6}$$

where the ellipsis denotes higher order commutators. But this is still not very helpful since $[A, B]$ is again an operator whose eigenstates we need to consider. This is where we use the following trick by inserting a complete set of intermediate states to

[1] *Hint:* Insert a complete set of energy eigenstates (which satisfy $H|n\rangle = E_n|n\rangle$) with $\mathbb{I} = \sum_n |n\rangle\langle n|$ and use their orthonormality property $\langle n|m\rangle = \delta_{nm}$ to show this.

get,

$$\langle x_f|U(t_f - t_i)|x_i\rangle = \langle x_f|e^{-iH(t_f-t_1-t_i+t_1)/\hbar}|x_i\rangle \tag{7}$$
$$=\langle x_f|e^{-iH(t_f-t_1)/\hbar}e^{-iH(t_1-t_i)/\hbar}|x_i\rangle \tag{8}$$
$$=\int dx_1\langle x_f|e^{-iH(t_f-t_1)/\hbar}|x_1\rangle\langle x_1|e^{-iH(t_1-t_i)/\hbar}|x_i\rangle \tag{9}$$

We can repeat this process any number of times by inserting more intermediate states. Let us say we insert $N-1$ intermediate states at times $t_1, t_2, \cdots, t_{N-1}$ such that $t_f - t_i = N\epsilon$. Then we can write,

$$\langle x_f|U(t_f - t_i)|x_i\rangle = \int dx_1 \cdots \int dx_{N-1}\langle x_f|e^{-iH\epsilon/\hbar}|x_{N-1}\rangle \cdots \langle x_1|e^{-iH\epsilon/\hbar}|x_i\rangle \tag{10}$$

Inserting the Hamiltonian from Eq. (5) and using the BCH formula to split the exponentials, neglecting higher order commutators ($\mathcal{O}(\epsilon^2)$), we can write,

$$\begin{aligned}\langle x_{i+1}|U(t_{i+1} - t_i)|x_i\rangle &= \langle x_{i+1}|e^{-i\frac{p^2}{2m}\epsilon/\hbar}e^{-iV(x)\epsilon/\hbar}|x_i\rangle + \mathcal{O}(\epsilon^2)\\ &= \int dp\ \langle x_{i+1}|e^{-i\frac{p^2}{2m}\epsilon/\hbar}|p\rangle\langle p|e^{-iV(x)\epsilon/\hbar}|x_i\rangle\\ &= \int dp\ e^{-i\frac{p^2}{2m}\epsilon/\hbar}e^{-iV(x_i)\epsilon/\hbar}\langle x_{i+1}|p\rangle\langle p|x_i\rangle\\ &= \int dp\ e^{-i\frac{p^2}{2m}\epsilon/\hbar}e^{-iV(x_i)\epsilon/\hbar}\frac{1}{2\pi\hbar}e^{ip(x_{i+1}-x_i)/\hbar}.\end{aligned} \tag{11}$$

In the above, we have used the position representation of the momentum eigenstates, $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}$. Note that the integral over p is not well-defined because of being highly oscillatory[2]. However, a small rotation of the infinitesimal time step ϵ into the complex plane, $\epsilon \to \epsilon - ia$ with $a > 0$, makes the integral convergent. This gives us the factor $e^{-a\frac{p^2}{2m}\epsilon/\hbar}$ which damps the oscillations and allows us to evaluate the integral. Now we complete the square in the exponent, carry out the 1D Gaussian integral and set $a \to 0$ to get,

$$\langle x_{i+1}|U(t_{i+1} - t_i)|x_i\rangle = \sqrt{\frac{m}{2\pi i\hbar\epsilon}}e^{\frac{i}{\hbar}\epsilon\left[\frac{m}{2}\left(\frac{x_{i+1}-x_i}{\epsilon}\right)^2 - V(x_i)\right]}. \tag{12}$$

Considering the limit $\epsilon \to 0$, one can recognize the term in the square brackets of the exponential as the Lagrangian of the theory, that when integrated over time gives us the action as,

$$\begin{aligned}S[x] &= \int dt\left[\frac{m}{2}(\partial_t x)^2 - V(x)\right]\\ &= \lim_{\epsilon\to 0}\sum_i \epsilon\left[\frac{m}{2}\left(\frac{x_{i+1}-x_i}{\epsilon}\right)^2 - V(x_i)\right]\end{aligned} \tag{13}$$

Finally, inserting Eq. (12) into Eq. (10) and taking the *time continuum limit*, we arrive at the *path integral* or the *functional integral* representation of the transition amplitude given by,

$$\langle x_f|U(t_f - t_i)|x_i\rangle = \int \mathcal{D}x\ e^{iS[x]/\hbar} \tag{14}$$

[2] You are encouraged to check this integral's Lebesgue and square integrability.

where the *path integral measure* in this simple quantum mechanical case is defined as,

$$\int \mathcal{D}(x) = \lim_{\epsilon \to 0} \sqrt{\frac{m}{2\pi i \hbar \epsilon}}^{N-1} \int dx_1 \int dx_2 \cdots \int dx_{N-1}. \tag{15}$$

This represents an integration over all possible paths at intermediate time steps, weighted by an oscillatory phase $e^{iS[x]/\hbar}$. In the *classical limit*, where $S[x] \gg \hbar$, or in standard terms $\hbar \to 0$, the largest contribution to the path integral comes from the paths corresponding to the solutions to the classical equations of motion.

3 Euclidean path integral

The original motivation to study purely-imaginary (Euclidean) time path integral comes from recognizing the connection to statistical mechanics. If one considers the *partition function* in quantum statistical mechanics,

$$Z = \text{Tr} e^{-\beta H}, \tag{16}$$

where H is the Hamiltonian and $\beta = \frac{1}{T}$ (we set $k_B = 1$) is the inverse temperature, one immediately recognizes a mathematical connection to the real time transition amplitude $\langle x | U(t_f - t_i) | x \rangle$, if we identify,

$$\beta = \frac{i}{\hbar}(t_f - t_i). \tag{17}$$

With this identification, one can divide the Euclidean time interval into infinitesimal time steps $\beta = Na/\hbar$, and following the same procedure as done in the previous section: inserting intermediate position eigenstates, using the BCH formula to ignore commutators of p^2 and $V(x)$ and carrying out the - now already *well defined* Gaussian integral, we get the Euclidean result for Eq. (14) as,

$$Z = \int \mathcal{D}x \; e^{-S_E[x]/\hbar}. \tag{18}$$

Notice that the subscript E on the Euclidean action is *not just notational*! Our excursion to imaginary time changed the form of the action to,

$$S_E[x] = \int d\tau \; \left[\frac{m}{2} (\partial_\tau x)^2 + V(x) \right], \tag{19}$$

where $\tau = it$. Furthermore, because the partition function in Eq. (16) includes a trace over all states $\int dx \; \langle x | e^{-\beta H/\hbar} | x \rangle$, the measure also includes an extra term giving us ,

There is an exercise related to counting the number of integrals one needs for a given field theory on a given spacetime lattice!

$$\int \mathcal{D}(x) = \lim_{a \to 0} \sqrt{\frac{m}{2\pi \hbar a}}^{N} \int dx \int dx_1 \int dx_2 \cdots \int dx_{N-1}. \tag{20}$$

After having formally defined the Euclidean path integral for the quantum mechanical system, we need to upgrade the notation to field theory notation. In a field theory the position x just becomes a label defining the value of a field in spacetime. Because of this we need to consider the derivatives of this field with respect to not just time but also space. This gives us the following dictionary between what we studied above for a non-relativistic quantum mechanical particle to a quantum field theory governing a field $\phi(x)$,

$$\begin{aligned} t &\longrightarrow x=(t,\vec{x}) \\ x &\longrightarrow \phi(x) \\ S[x]=\int dt\ L(x,\partial_t x) &\longrightarrow S[\phi]=\int d^4x\ L(\phi,\partial_\mu\phi) \\ \mathcal{D}x=\prod_i\int dx_i &\longrightarrow \mathcal{D}\phi=\prod_x\int d\phi(x). \end{aligned} \tag{21}$$

Now that we have established a connection between the Euclidean path integral of a quantum field theory and the partition function of a classical statistical-mechanical system, we can exploit this correspondence to compute thermal expectation values of observables. This is possible because of the interpretation of $e^{-S_E[\phi]/\hbar}$ as the probability weight for a given *configuration* $\phi(x)$ — precisely analogous to the Boltzmann weight $e^{-\beta E}$ in classical statistical mechanics . Then the expectation value of an arbitrary observable $O(\phi)$ is formally given as,

Computing such expectation values is the primary objective in lattice QCD research, especially the area focused on the QCD phase diagram. For further reading see [3]

$$\langle O\rangle=\frac{\int\mathcal{D}\phi\ O(\phi)\ e^{-S_E[\phi]/\hbar}}{\int\mathcal{D}\phi\ e^{-S_E[\phi]/\hbar}} \tag{22}$$

From the more field-theoretic point of view, one can also compute two-point (or higher N-point) functions from the Euclidean path integral in analogy with what we are used to from the perturbative formulation of quantum field theories. In this way one can construct the Euclidean Green's functions that contain information about the particle spectrum. For example, the following two-point function (Green's function),

$$G_E(\vec{x},\tau;\vec{x}',0)=\langle\phi(\vec{x},\tau)\phi(\vec{x}',0)\rangle=\frac{\int\mathcal{D}\phi\ \phi(\vec{x},\tau)\phi(\vec{x}',0)\ e^{-S_E[\phi]/\hbar}}{\int\mathcal{D}\phi\ e^{-S_E[\phi]/\hbar}}, \tag{23}$$

is of very high importance in lattice studies as this quantity contains information about the particle spectrum of the theory. The decay of this two-point function gives us information on the mass gap of the theory.

This brings us to the problem of dealing with divergences - computing Green's functions (both in Minkowski or Euclidean spacetime) are tricky because of short-distance or large-momentum (called Ultra-violet or UV) divergences . If you try to directly compute this you will simply get a bunch of infinities! But this should be confusing because we know that quantities like Eq. (23) contain information on physical quantities like masses of particles - which are certainly not infinite. Thus we are forced to introduce some regulators like a UV-cutoff which render the computation of these quantities finite and, very importantly, we need to understand how these quantities change as we remove the regulator while keeping the Green's function finite.

There is an exercise related to UV divergence simple quantities and how a lattice tackles this

Now we are ready to introduce regularization scheme that is in the title of this course-Lattice field theory.

4 Lattice (scalar) field theory

In this part of the lecture we will see how introducing a spacetime lattice regulates the formal path integral measure and how it removes the above-mentioned UV divergences. For this we will consider the case of the scalar ϕ-field theory. Note that this is meant to be a *warm-up* for the rest of the course - in general we will not be dealing with scalar fields - but with the fermion and gauge fields in the subsequent lectures.

To begin we consider the (continuum) Euclidean action corresponding to the scalar field ϕ field in 4 spacetime dimensions (from now on x denotes the 4 vector (x_1, x_2, x_3, x_4)),

$$S_E[\phi] = \frac{1}{2}\int d^4x \; \phi(x) \left(-\Box + M^2\right) \phi(x), \tag{24}$$

Using integration by parts, one can show that this is the same as $S_E[\phi] = \frac{1}{2}\int d^4x \; \partial_\mu\phi(x)\partial_\mu\phi(x) + M^2\phi^2(x)$ - a form that we will use in the next lecture.

where $\Box$ denotes the 4-dimensional Laplacian $\sum_{\mu=1}^{4} \partial_\mu\partial_\mu$. then the corresponding partition function is (setting $\hbar = 1$ from now onward),

$$Z = \int \prod_x d\phi(x) \; e^{-S[\phi]}. \tag{25}$$

In order to see how these quantities look like on a discrete spacetime lattice, consider the two-dimensional version in Fig. 1. This gives us the following prescription :

In the exercise sheet you will have the opportunity to derive the expressions in momentum space for the derivatives.

$$\begin{array}{rcl}
x_\mu & \longrightarrow & n_\mu a\,, \\
\phi(x) & \longrightarrow & \phi(na)\,, \\
\int d^4x & \longrightarrow & a^4 \sum_n\,, \\
\partial_\mu\phi(x) & \longrightarrow & \Delta^f_\mu = \frac{\phi(na+\hat{\mu}a)-\phi(na)}{a}\,, \\
\partial_\mu\phi(x) & \longrightarrow & \Delta^b_\mu = \frac{\phi(na)-\phi(na-\hat{\mu}a)}{a}\,, \\
\partial_\mu\phi(x) & \longrightarrow & \Delta^s_\mu = \frac{\phi(na+\hat{\mu}a)-\phi(na-\hat{\mu}a)}{2a}\,, \\
\mathcal{D}\phi & \longrightarrow & \prod_n \int d\phi(na).
\end{array} \tag{26}$$

We can immediately see from the Eq. (26), that there are many choices of constructing the lattice Laplacian operator $\Box_L$ - we can use the forward difference operators, the backward difference operators, the symmetric or a combination of any two of these. Since we are in any case summing over all spacetime points when computing the action, and we have periodic boundary conditions - some of the combinations will give the same result. For this lecture we choose the conventional choice of,

$$\Box_L\hat{\phi}(na) = \Delta^b_\mu\Delta^f_\mu = \frac{1}{a^2}\sum_\mu \left(\phi(na+\hat{\mu}a) + \phi(na-\hat{\mu}a) - 2\phi(na)\right)\,. \tag{27}$$

In order to write down the full lattice action for the scalar field theory, we need to consider the dimensionality of the lattice parameters like the field value $\phi(na)$ and the lattice mass M. We need to maintain the dimensionlessness of the Euclidean lattice action. Looking at Eq. (13) and Eq. (27), we get

There is an exercise related to this!

$$\begin{array}{rcl}
\hat{\phi}_n & = & a\phi(na) \\
\hat{M} & = & aM \\
\hat{\Box} & = & a^2\Box_L\,,
\end{array} \tag{28}$$

defining the dimensionless lattice quantities with hats and removing the label L from the box operator. Using this one can re-write the Euclidean lattice action as ,

The factor $2d$ ($= 8$ in 4 dimensions) comes from the diagonal part of $-\hat{\Box}$, summing over all directions.

$$S_E = -\frac{1}{2}\sum_{n,\hat{\mu}} \hat{\phi}_n \hat{\phi}_{n+\hat{\mu}} + \frac{1}{2}\left(8 + \hat{M}^2\right)\sum_n \hat{\phi}_n \hat{\phi}_n. \tag{29}$$

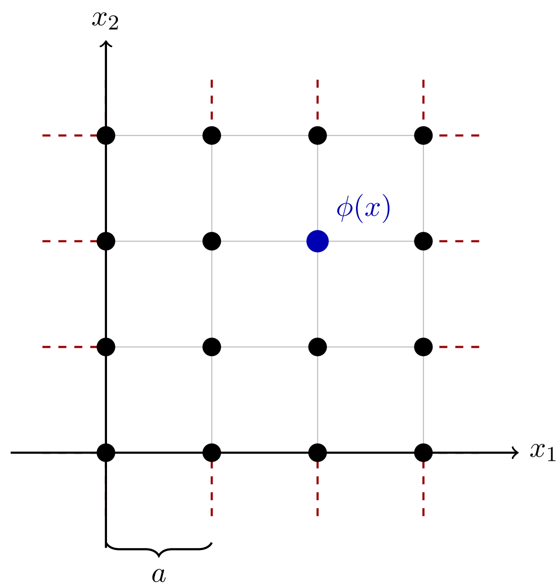

Figure 1: A 4×4 two-dimensional lattice with spacing a and periodic boundary conditions. Matter fields live on lattice sites as indicated by $\phi(x)$. Dashed red lines indicate that opposite edges are identified: $\phi(x + L\hat{\mu}) = \phi(x)$.

Before moving on to seeing how this spacetime discretization helps in considering observables coming from the path integral - we need to discuss what happens to the allowed values of momentum when one discretizes space. The easiest way to see this is to consider an infinite line segment and write down the equations for the Fourier transform and the inverse Fourier transform of a function along this line,

$$\begin{aligned} \tilde{f}(p) =& \frac{1}{2\pi}\int_{-\infty}^{\infty} dx\; f(x)\, e^{-ipx} \\ f(x) =& \int_{-\infty}^{\infty} dp\; \tilde{f}(p)\, e^{ipx}\,. \end{aligned} \tag{30}$$

In order to see what happens to the momentum when we discretize space - we need to consider two limits. First consider the case where the line segment is not continuous but is made up of discrete points with spacing a, but the points still span infinity. In this case the first equation in Eq. (30) becomes,

There is an exercise related to this!

$$\tilde{f}(p) = \frac{a}{2\pi}\sum_{n=-\infty}^{\infty} f(na)\, e^{-inpa}\,. \tag{31}$$

This immediately tells us that $\tilde{f}(p)$ is periodic with period $2\pi/a$! This means that we can limit the integration in momenta over one period - called the Brillouin Zone

$[-\pi/a, \pi/a)$ and the integration limits for any momentum integral is now limited! This is how a discrete spacetime lattice regularizes the theory - by removing the UV-divergences arising due to the presence of very short-distance scales or very large momenta.

However, we are not done yet as the lattice also has a *finite extent*. This will further restrict the allowed momenta values on the lattice! For this we remind ourselves that our theory lives on a lattice with periodic boundary conditions. For the line segment, this means $f(x+L) = f(x)$. Combining this with the second equation in Eq. (30), one finds that the finite extent of the line segment limits the number of allowed momenta to be $p = \frac{2\pi n}{L}$ for $n \in [-L/2a, L/2a)$ (recall that $L = Na$).

You will see a more detailed treatment of lattice momenta in Nico Dichter's treatment of fermions on the lattice on 24.06.

. . . .

Finally, to compute the two-point function for the theory, we rewrite the action in Eq. (29) in the form,

$$S_E = \frac{1}{2} \sum_{n,m} \hat{\phi}_n K_{nm} \hat{\phi}_m \tag{32}$$

where the matrix elements K_{nm} are given by reading off what is *sandwiched* between the ϕ's in Eq. (29) and considering only positive directions,

$$K_{nm} = -\sum_{\mu>0} (\delta_{n+\hat{\mu},m} + \delta_{n-\hat{\mu},m} - 2\delta_{mn}) + \hat{M}^2 \delta_{mn} \,. \tag{33}$$

Using this we can write the lattice partition function as (compare to Eq. (25)),

$$Z = \int \prod_l d\phi_l \ \exp\left(-\frac{1}{2} \sum_{n,m} \hat{\phi}_n K_{nm} \hat{\phi}_m \right) \tag{34}$$

In the field theory treatment of N-point functions, one typically introduces a *source term* for the fields, $J_n \phi_n$, in order to get a generating functional,

You can find more about this type of calculation and more information on generating functionals in classic QFT textbooks like[4].

$$Z[J] = \int \prod_l d\phi_l \ \exp\left(-\frac{1}{2} \sum_{n,m} \hat{\phi}_n K_{nm} \hat{\phi}_m + J_n \phi_n \right) \,. \tag{35}$$

Because of the Gaussian-like nature of the individual terms in the exponent, the integral(s) in Eq. (35) can be computed easily[3] to get,

$$Z[J] = \frac{1}{\sqrt{K}} \exp\left(\sum_{n,m} J_m K^{-1}_{m,n} J_n \right) \tag{36}$$

Now we just have to use one more fact about generating functionals-that their derivatives with respect to J_l give rise to N-point functions and then set $J = 0$. Since we are interested in the two-point function, after some algebra we get,

$$\langle \hat{\phi}_n \hat{\phi}_m \rangle = \frac{\delta Z[J]}{\delta J_n \delta J_m}|_{J=0} = K^{-1}_{m,n} \tag{37}$$

Hence, all that is left to do is find the inverse of the matrix in Eq. (33)! For this we will use two ingredients: (i) use the fact that $\sum_l K_{m,l} K^{-1}_{l,n} = \delta_{m,n}$ and, (ii) do the computation in momentum basis using the lattice version of the Fourier transform

[3] This will be an exercise in the *next sheet*!

as discussed before. One can construct the Fourier transform of Eq. (33) in a straightforward manner to get,

$$K(\hat{p}) = 4\sum_{\mu=1}^{4} \sin^2(\hat{p}_\mu/2) + \hat{M}^2 \,. \tag{38}$$

Plugging the inverse of this equation into the momentum space representation of the matrix inverse identity, one gets

It would be a nice exercise to show this!

$$K^{-1}_{nm} = \langle \hat{\phi}_n \hat{\phi}_m \rangle = \int_{-\pi,\pi} \frac{d^4\hat{p}}{(2\pi)^4} \, \frac{e^{i\hat{p}\cdot(n-m)}}{4\displaystyle\sum_{\mu=1}^{4} \sin^2\left(\frac{\hat{p}_\mu}{2}\right) + \hat{M}^2} \,. \tag{39}$$

With this we have our first well-defined and finite lattice quantity - the two-point Green's function written completely in terms of lattice regulated variables like the dimensionless mass $\hat{M}$, lattice sites m, n, lattice momenta $\hat{p}$,

$$G(n, m; \hat{M}) \equiv \langle \hat{\phi}_n \; \hat{\phi}_m \rangle = K^{-1}_{nm} \,. \tag{40}$$

We finally come to the *formally* last part of the procedure - removing the lattice structure while keeping, for example, the Green's functions finite. This is something we will discuss in detail in a future lecture and forms the most important step of a lattice computation. For now, because we have a very simple free scalar field theory, the continuum limit of the Green's function in Eq. (39) is easy to show analytically, by demanding that we take the limit $a \to 0$ in such a way that the physical mass of the scalar and the lattice volume remain fixed while removing the cutoff. This means that the bare lattice quantities necessarily change as a function of a!

However, if you start reading LFT research papers you will see that it is a very rare finding that someone actually takes the continuum limit in their calculations - most lattice physicists just simulate one large and fine lattice and publish their results

Re-introducting the lattice structure like the spacing in Eq. (40), we get,

$$\langle \phi(x)\phi(y) \rangle \equiv \lim_{a\to 0} \frac{1}{a^2} G\left(\frac{x}{a}; \frac{y}{a}; Ma\right) \,, \tag{41}$$

where we know that the left-hand-side of the equation is a finite quantity and in order to make the right-hand-side finite, we require

$$\lim_{a\to 0} G\left(\frac{x}{a}; \frac{y}{a}; Ma\right) \to 0 \,. \tag{42}$$

This is seen by replacing the dimensionless quantities by the dimensionful versions and introducing the accompanying factors of a to get

$$G\left(\frac{x}{a}; \frac{y}{a}; Ma\right) = a^2 \int_{-\pi/a}^{\pi/a} \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot(x-y)}}{\bar{p}_\mu^2 + M^2} \,, \tag{43}$$

where $\bar{p}_\mu$ is given by

$$\bar{p}_\mu = \frac{2}{a} \sin\left(\frac{p_\mu a}{2}\right) \,. \tag{3.19b}$$

Because the momentum integral in Eq. (43) is limited to the finite interval $[-\pi/a, \pi/a]$, high-momentum modes (of order $1/a$) are excluded and the integrand is controlled by momenta that are tiny compared with the inverse lattice spacing. In this regime $\bar{p}_\mu$ reduces to the ordinary continuum momentum p_μ; letting the lattice spacing a go to zero then reproduces the usual continuum propagator:

$$\langle \phi(x)\phi(y) \rangle = \int_{-\infty}^{\infty} \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot(x-y)}}{p^2 + M^2} \,. \tag{44}$$

This exercise was just to show that the lattice regularization is a valid procedure and with the correct procedure we recover our usual continuum physics. Hence, lattice field theories offer us a way to go beyond perturbation theory and compute finite quantities while giving a practical meaning to the path integral measure.

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